

SAYAN GHORAI

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[LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

M.Tech candidate in Artificial Intelligence and Data Science with hands-on experience building and deploying end-to-end AI/ML systems — including deep learning, generative AI, and NLP pipelines — across three production-style projects. Strong foundation in Python, Data Structures & Algorithms, DBMS, and software engineering practices, with additional exposure to enterprise application development through a Salesforce internship. Seeking AI/ML Engineer or Software Engineer roles where I can apply model development, MLOps, and full-stack deployment skills to solve real-world problems.

TECHNICAL SKILLS

- **Programming Languages:** Python, C
- **Core CS Concepts:** Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems
- **AI / Machine Learning:** Machine Learning, Deep Learning, CNN, RNN, LSTM, GAN, Random Forest, SVM
- **Generative AI:** LoRA Fine-Tuning, Vision-Language Models, Semantic Retrieval, Multimodal AI
- **Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-learn, Transformers, PEFT, SBERT, OpenCV, Pandas, NumPy
- **Database:** MySQL
- **MLOps & Deployment:** Docker, Kubernetes, Kubeflow, MLflow, Streamlit
- **Tools & Platforms:** Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Kaggle

PROJECTS

GenAI Indian Tourist Place Recommendation System | [GitHub](#) | [Live Demo](#)

- Engineered a hybrid retrieval engine combining TF-IDF lexical search with Sentence-BERT (all-MiniLM-L6-v2) semantic embeddings across a dataset of 1.5M+ reviews spanning 15,000+ tourist places and 1,782 Indian cities.
- Built city-detection and intent-detection modules to parse natural-language travel queries into structured city, intent, and tag concepts, with automatic tagging and rating x review-count confidence scoring for ranking.
- Designed an explainable recommendation layer that surfaces the metadata behind each suggestion, and deployed the system as a modular Streamlit application with a reusable backend/CLI architecture.

Plant Disease Classification | [GitHub](#) | [Live Demo](#)

- Built an ensemble deep learning model (ResNet50 + ShuffleNetV2) in PyTorch for 38-class plant disease classification across 14 plant species, trained on 87,000+ leaf images (70,295 train / 17,572 validation).
- Achieved 99.13% validation accuracy and 99.11% test accuracy; implemented Grad-CAM explainability and OpenCV-based leaf segmentation and validation to improve prediction robustness and interpretability.
- Optimized the model for deployment via quantization and TorchScript, and shipped a Streamlit web app with confidence scoring and disease cause/treatment/prevention recommendations.

Multilingual Image Captioning Assistant | [GitHub](#) | [Live Demo](#)

- Fine-tuned the Florence-2 vision-language model using LoRA (parameter-efficient fine-tuning) on 50,000 COCO image-caption pairs to generate natural-language image captions under limited GPU memory.
- Built an end-to-end multimodal pipeline — image captioning to Bengali translation to text-to-speech synthesis — to support accessibility use cases.
- Deployed an interactive Streamlit application integrating computer vision, NLP, and speech technologies into a single multimodal assistant.

INTERNSHIP EXPERIENCE

Salesforce Developer Virtual Internship — SmartBridge / SmartInternz (AICTE)

Nov 2023 – Jan 2024

- Completed Salesforce Trailhead modules covering Fundamentals, Organizational Setup, Process Automation, Flows & Security, and Apex Testing & Debugging; earned 3 Superbadges (Apex Specialist, Process Automation Specialist, Developer Super Set).
- Built automation workflows using Apex programming, integrated APIs, and developed Lightning Web Components, gaining exposure to enterprise cloud application development practices.

EDUCATION

VIT Bhopal University — M.Tech, Artificial Intelligence and Data Science

2024 – 2026 | CGPA: 7.59/10

Haldia Institute of Technology — B.Tech, CSE (AI & Machine Learning)

2020 – 2024 | CGPA: 8.53/10

CERTIFICATIONS

- Salesforce Developer Virtual Internship Certificate — SmartBridge / SmartInternz, in partnership with AICTE (Feb 2024)
- Python for Beginners Course In-Depth — Udemy (Feb 2026)